

A Sparrow's Song: Balancing Art and Innovation for an Economic CG Production

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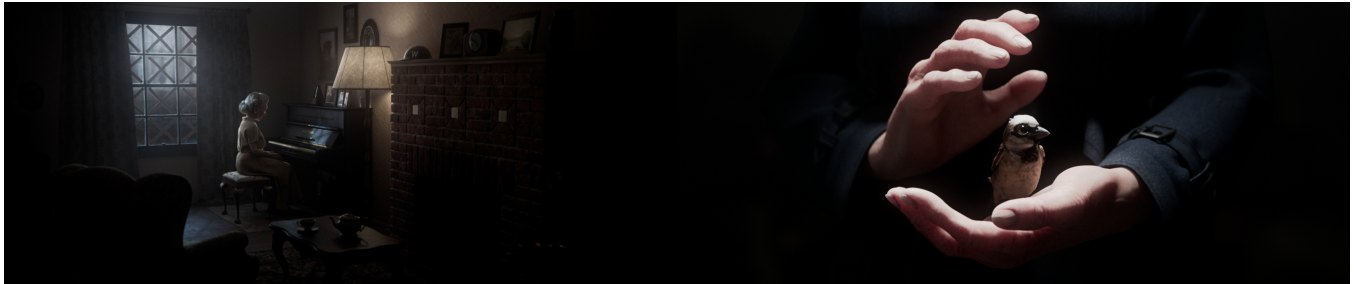


Figure 1: "A Sparrow's Song" film stills

Abstract

Inspired by a true story, a widowed air raid warden in the midst of World War II struggles to overcome grief and rediscover joy in her life—until she finds a dying sparrow she hopes to save. The 10-minute full CG animated short film, "A Sparrow's Song", created as a diploma project at the Animationsinstitut of Filmakademie Baden-Württemberg, offers a unique perspective on life in war-torn London during the Blitz. Combining modern production technologies with traditional CG workflows, we brought this distinctive story to life. This paper details key lessons learned, addressing the challenges and creative solutions in delivering a high-quality animated short within the constraints of a student production.

CCS Concepts

• **Applied computing** → **Media arts**; • **Computing methodologies** → **Animation**; **Rendering**.

Keywords

CG Production, Animated Short Film, Rendering, Optimization

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1 Introduction

"A Sparrow's Song" is a CG-animated short, following a widowed air raid warden who finds a dying sparrow in WWII. The film was created by Animationsinstitut students and recent graduates over a span of two and a half years. We wanted to create a compelling emotional story, while pushing the boundaries of technological and creative innovation. A linear timeline with fixed milestones enabled us to stay agile with minimal dependencies. This structure allowed us to adjust goals and develop a scalable production pipeline, starting from a minimum viable product and adding features as needed.

2 Story and Concept

After an initial script, inspired by a true story, we defined the film's look and feel through concept art. *Chiaroscuro* lighting, emphasizing contrast between light and dark, was the guiding principle in creating our images. Each shot was crafted to stand for itself, making every frame a painting. From first concept to final render, we concentrated on what was important to the story, discarding unnecessary details. Consistent timing and staging from animatic to final edit allowed asset and layout artists to create only the on-screen assets. Meanwhile, the animation department could focus on the movements essential for storytelling.

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3 Assets

Our asset work was split into three categories: characters, props, and environments. Characters were designed with a stylized, yet realistic approach. As the sparrow's health improves throughout the film, multiple versions of its body and feather system were created. Procedural deformation proved sufficient to create realistic feather movement and minimized the need for simulation. By concealing human faces behind masks and mainly showing the widow's emotions through her hands, we minimized time-consuming facial animation. A shared base geometry and rig made it quick to adapt blend shapes and animations to the different characters. Prop creation was made efficient by reviewing assets in shot context, and a texture-driven shading approach allowed one artist to shade over 200 assets. For environments, the living room set dressing focused on fixed camera angles; in contrast, the modular air raid shelter was broadly dressed to accommodate a diverse set of shots, while a single generalist created the London establisher.

4 Animation

We opted for minimalistic splined animation, focusing only on what was essential for the story. Animating the sparrow's feathers, crucial to its silhouette, required a complex rig with blend shapes for different poses to create the challenging opening and closing animation. For the realistic piano animation we tested motion capture, but it was too inaccurate for close-up shots. Instead, the musician's performance was recorded from multiple angles as reference, exported as MIDI, and used in a Maya Python script to generate the base animation for the animators.

5 Pipeline and Crowds

Our goal was to build a workflow that was not reliant on a fully set-up pipeline, allowing us to adapt to new challenges and proceed with production quickly. The team found it beneficial to use proven methods that artists were comfortable with over cutting-edge tools, so we decided to use an Alembic-based pipeline instead of USD. The motionless nature of the frightened crowd allowed us to take some shortcuts. We created 10 background characters with shared idle animations and developed a hybrid cloth approach, simulating cloth on static poses before skinning to idle animation, rather than wrapping or simulating all characters. Crowd proxies were placed in Maya via MASH and exported as USD instances to Katana¹.

6 Lighting and Compositing

For shading, lighting, and rendering, we used Katana together with Arnold. Its ability to handle large amounts of geometry, along with its *templating* and *multi-shot workflow*, allowed us to light over 70 shots with just two lighting artists over a period of about four months. However, this came with a steeper learning curve and some integration challenges. To achieve an analog WWII era film look, the cinematography was founded on real vintage lenses and minimal camera animation. Realistic lens flares and glows were created using our research project, "Capturing Light with Robots: A Novel Workflow for Reproducing Realistic Lens Flares" [Maurer 2024]. Custom lens kernels simulated depth of field in compositing,

and a particle-based system emulated film grain, which, combined with *gate weave* and *halation*, added rich texture to the images. In addition to creating a color grade in SDR (standard dynamic range), we were the first project at Filmakademie to master a film in Dolby Vision HDR² (high dynamic range). This enhances visuals with wider contrast and color gamut and speeds up multi-deliveries, but requires special attention in compositing and quality control.

7 Rendering and Optimization

At Animationsinstitut, we prioritize green production initiatives, such as minimizing our energy consumption. We evaluated real time rendering for our production, as it allows for more iterations, similar appearance and a significant reduction of energy consumption compared to offline rendering, as shown in the Animationsinstitut talk "Sustainable VFX - An Educational Challenge" [Helzle et al. 2024]. We ultimately decided against it due to challenges in rendering hair, skin, and feathers, as well as issues with achieving realistic global illumination. The production utilized Arnold's GPU render in combination with real-time techniques like hair cards for crowds and fog cards to reduce render times (10 - 30 min per frame). The static cameras (allowing to render backgrounds only once), minimal re-renders and denoising helped the film stay within its 8000 kWh energy consumption target on the in-house render farm, roughly half of a typical final year 3D animated short at Animationsinstitut.

8 Conclusion

Looking back on the two and a half years of production, we are very excited about the results we have achieved. Although we are mostly happy with our pipeline, after the latest advancements in technology and tools, we would reconsider a USD-based workflow. Some of the more classical CG approaches, like the crowd scattering, could be replaced by their procedural counterparts. Our approach to world design, lighting concepts, and minimal animation played a crucial role in making this film possible. We prioritized choosing the "right tool for the job" rather than always opting for the latest technology. The mindset of letting go of perfectionism is also something we want to maintain in future projects. Thanks to the clear vision, the motivated team and the scalable production, we exceeded our expectations of what is possible.

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¹<https://www.foundry.com/products/katana>

²<https://www.dolby.com/technologies/dolby-vision/>